1 Basic Definitions

Mathematical modeling often relates a quantity to its derivatives in the form on an equation.

Definition 1.1 (Differential equation)**.** A *differential equation* is an equation involving an unknown function $y = f(x)$ and one or more of its derivatives.

Example 1.2. Let $A(t)$ denote the amount of some quantity at time t. Suppose the rate of change of this quantity at time t is proportional to the amount of the quantity present at that time. We model this by the differential equation

$$A' = kA,$$

for some constant k.

Definition 1.3 (Solution)**.** A *solution* to a differential equation is a function $y = f(x)$ that satisfies the differential equation when f and its derivatives are substituted into the equation.

Example 1.4. The function $A(t) = e^{kt}$ is a solution to the differential equation $A' = kA$ since $A'(t) = \frac{d}{dt}e^{kt} = ke^{kt} = kA(t)$ for all t.

1. Show that $A(t) = 0$ is also a solution to the differential equation $A' = kA$. This shows that the solution to a differential equation need not be unique.

2. Show that $y = e^{-3x} + 2x + 3$ is a solution to the differential equation $y' + 3y = 6x + 11$.

3. Show that $y = 3e^{x} - 4e^{2x} + 2e^{-2x}$ is a solution to the differential equation $y'' - 3y' + 2y = 24e^{-2x}$.

Definition 1.5 (Order)**.** The *order* of a differential equation is the highest order of any derivative of the unknown function that appears in the equation.

4. What is the order of each of the following differential equations?

   (a) $y' - 4y = x^2 - 3x + 4$

   (b) $x^2y''' - 3xy'' + xy' - 3y = \sin(x)$

   (c) $\frac{4}{x}y^{(4)} - \frac{6}{x^2}y'' + \frac{12}{x^4}y = x^3 - 3x^2 + 4x - 12$

2 General and Particular Solutions

Consider the differential equation $y' = 2x$. Clearly, any antiderivative of $2x$ is a solution, that is

$$y = \int y'dx = \int 2xdx = x^2 + C$$

is a solution for any real number C since $\frac{d}{dx}(x^2 + C) = 2x$, no matter the value of C. The set $\{x^2 + C : C \in \mathbb{R}\}$ is called the *general solution* to the differential equation $y' = 2x$. A graph of some of these solutions is shown below:

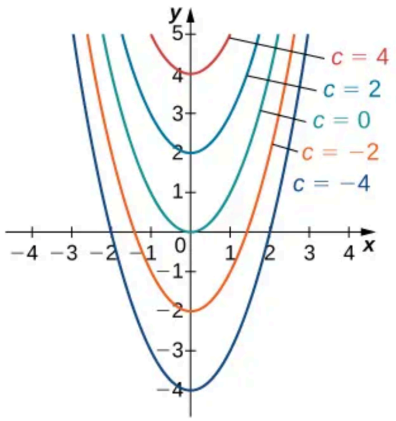


Figure 1: Family of solutions to the differential equation $y' = 2x$.

If we fix a value of C, then we get a *particular solution* of the differential equation. A particular solution can often be identified if we are given additional information about the problem. For instance, if we want the particular solution to the differential equation $y' = 2x$ which passes through the point $(2,7)$, then solving $7 = (2)^2 + C$ gives $C = 3$, which singles out the particular solution $y = x^2 + 3$.

3 Initial-Value Problems

As we have just seen in the example above, usually a given differential equation has infinitely many solutions. To choose one solution, more information is needed. In many situaitons we need to find the particular solution that satisfies a condition of the form $y(t_0) = y_0$, where t_0 is a fixed value of the independent variable, and y_0 is a fixed value of the dependent variable, which is exactly what we did in the example above. The ordered pair (t_0, y_0) is an example of an *initial condition*, and the problem of finding a solution of a differential equation that satisfies the initial condition is called an *initial-value problem*. Geometrically, when we impose an initial condition, we look at the family of solution curves and pick out the one that passes through the point (t_0, y_0). For a higher-order differential equation, we also potentially need to specify $y'(t_0)$, $y''(t_0)$, etc. The general rule is that the number of initial values needed for an initial-value problem is equal to the order of the differential equation.

5. Verify that the function $y = 2e^{-2t} + e^t$ is a solution to the initial-value problem

$$y' + 2y = 3e^t, \;\; y(0) = 3.$$

6. Solve the initial-value problem $y' = 3e^x + x^2 - 4, \quad y(0) = 5.$

4 Application: Free fall

The velocity of an object falling under the influence of gravity near the surface of the Earth is modeled by the differential equation

$$v'(t) = -g,$$

where $g \approx 9.8 m/s^2$ (assuming that all other forces, including air resistance, can be neglected).

7. A baseball is thrown upward from a height of 3m about Earth's surface with an initial velocity of 10 m/s, and the only force acting on it is gravity.

   (a) Find the velocity $v(t)$ of the baseball at time t.

   (b) What is its velocity after 2 seconds?